```
mysql> SELECT * FROM employees;
+-------------+-------------+-----------+---------------+--------+
| employee_id | first_name  | last_name | device_serial | salary |
+-------------+-------------+-----------+---------------+--------+
|           1 | John        | Smith     | ABC123        |  60000 |
|           2 | Jane        | Doe       | DEF456        |  65000 |
|           3 | Bob         | Johnson   | GHI789        |  70000 |
|           4 | Sally       | Fields    | JKL012        |  75000 |
|           5 | Michael     | Smith     | MNO345        |  80000 |
|           6 | Emily       | Jones     | PQR678        |  85000 |
|           7 | David       | Williams  | STU901        |  90000 |
|           8 | Sarah       | Johnson   | VWX234        |  95000 |
|           9 | James       | Brown     | YZA567        | 100000 |
|          10 | Emma        | Miller    | BCD890        | 105000 |
|          11 | William     | Davis     | EFG123        | 110000 |
|          12 | Olivia      | Garcia    | HIJ456        | 115000 |
|          13 | Christopher | Rodriguez | KLM789        | 120000 |
|          14 | Isabella    | Wilson    | NOP012        | 125000 |
|          15 | Matthew     | Martinez  | QRS345        | 130000 |
|          16 | Sophia      | Anderson  | TUV678        | 135000 |
|          17 | Daniel      | Smith     | WXY901        | 140000 |
|          18 | Mia         | Thomas    | ZAB234        | 145000 |
|          19 | Joseph      | Hernandez | CDE567        | 150000 |
|          20 | Abigail     | Smith     | FGH890        | 155000 |
+-------------+-------------+-----------+---------------+--------+
20 rows in set (0.00 sec)
```

```
mysql> DESC employees;
+-----------------+-------------+------+-----+---------+-------+
| Field           | Type        | Null | Key | Default | Extra |
+-----------------+-------------+------+-----+---------+-------+
| employee_id     | int         | NO   | PRI | NULL    |       |
| first_name      | varchar(50) | YES  |     | NULL    |       |
| last_name       | varchar(50) | YES  |     | NULL    |       |
| device_serial   | varchar(15) | YES  |     | NULL    |       |
| salary          | int         | YES  |     | NULL    |       |
+-----------------+-------------+------+-----+---------+-------+
5 rows in set (0.04 sec)

mysql>
```

```
mysql> CREATE VIEW V1 as SELECT employee_id,salary FROM employees;
Query OK, 0 rows affected (0.05 sec)

mysql> DESC V1;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| employee_id | int  | NO   |     | NULL    |       |
| salary      | int  | YES  |     | NULL    |       |
+-------------+------+------+-----+---------+-------+
2 rows in set (0.00 sec)
```

```
mysql> INSERT INTO V1 VALUES(101,100001);
Query OK, 1 row affected (0.04 sec)

mysql> SELECT * FROM V1;
+-------------+--------+
| employee_id | salary |
+-------------+--------+
|           1 |  60000 |
|           2 |  65000 |
|           3 |  70000 |
|           4 |  75000 |
|           5 |  80000 |
|           6 |  85000 |
|           7 |  90000 |
|           8 |  95000 |
|           9 | 100000 |
|          10 | 105000 |
|          11 | 110000 |
|          12 | 115000 |
|          13 | 120000 |
|          14 | 125000 |
|          15 | 130000 |
|          16 | 135000 |
|          17 | 140000 |
|          18 | 145000 |
|          19 | 150000 |
|          20 | 155000 |
|         101 | 100001 |
+-------------+--------+
21 rows in set (0.00 sec)

mysql>
```

```
mysql> SELECT * FROM employees;
+-------------+------------+-----------+----------------+--------+
| employee_id | first_name | last_name | device_serial  | salary |
+-------------+------------+-----------+----------------+--------+
|           1 | John       | Smith     | ABC123         |  60000 |
|           2 | Jane       | Doe       | DEF456         |  65000 |
|           3 | Bob        | Johnson   | GHI789         |  70000 |
|           4 | Sally      | Fields    | JKL012         |  75000 |
|           5 | Michael    | Smith     | MNO345         |  80000 |
|           6 | Emily      | Jones     | PQR678         |  85000 |
|           7 | David      | Williams  | STU901         |  90000 |
|           8 | Sarah      | Johnson   | VWX234         |  95000 |
|           9 | James      | Brown     | YZA567         | 100000 |
|          10 | Emma       | Miller    | BCD890         | 105000 |
|          11 | William    | Davis     | EFG123         | 110000 |
|          12 | Olivia     | Garcia    | HIJ456         | 115000 |
|          13 | Christopher| Rodriguez | KLM789         | 120000 |
|          14 | Isabella   | Wilson    | NOP012         | 125000 |
|          15 | Matthew    | Martinez  | QRS345         | 130000 |
|          16 | Sophia     | Anderson  | TUV678         | 135000 |
|          17 | Daniel     | Smith     | WXY901         | 140000 |
|          18 | Mia        | Thomas    | ZAB234         | 145000 |
|          19 | Joseph     | Hernandez | CDE567         | 150000 |
|          20 | Abigail    | Smith     | FGH890         | 155000 |
|         101 | NULL       | NULL      | NULL           | 100001 |
+-------------+------------+-----------+----------------+--------+
21 rows in set (0.00 sec)

mysql>
```

```
mysql> SELECT * FROM V1;
+---------------+---------+
| employee_id | salary |
+---------------+---------+
|             1 |   60000 |
|             2 |   65000 |
|             3 |   70000 |
|             4 |   75000 |
|             5 |   80000 |
|             6 |   85000 |
|             7 |   90000 |
|             8 |   95000 |
|             9 |  100000 |
|            11 |  110000 |
|            12 |  115000 |
|            13 |  120000 |
|            14 |  125000 |
|            15 |  130000 |
|            16 |  135000 |
|            17 |  140000 |
|            18 |  145000 |
|            19 |  150000 |
|            20 |  155000 |
|           101 |  100001 |
+---------------+---------+
20 rows in set (0.00 sec)

mysql>
```

```
mysql> SELECT * FROM employees;
+-------------+-------------+-----------+---------------+--------+
| employee_id | first_name  | last_name | device_serial | salary |
+-------------+-------------+-----------+---------------+--------+
|           1 | John        | Smith     | ABC123        |  60000 |
|           2 | Jane        | Doe       | DEF456        |  65000 |
|           3 | Bob         | Johnson   | GHI789        |  70000 |
|           4 | Sally       | Fields    | JKL012        |  75000 |
|           5 | Michael     | Smith     | MNO345        |  80000 |
|           6 | Emily       | Jones     | PQR678        |  85000 |
|           7 | David       | Williams  | STU901        |  90000 |
|           8 | Sarah       | Johnson   | VWX234        |  95000 |
|           9 | James       | Brown     | YZA567        | 100000 |
|          11 | William     | Davis     | EFG123        | 110000 |
|          12 | Olivia      | Garcia    | HIJ456        | 115000 |
|          13 | Christopher | Rodriguez | KLM789        | 120000 |
|          14 | Isabella    | Wilson    | NOP012        | 125000 |
|          15 | Matthew     | Martinez  | QRS345        | 130000 |
|          16 | Sophia      | Anderson  | TUV678        | 135000 |
|          17 | Daniel      | Smith     | WXY901        | 140000 |
|          18 | Mia         | Thomas    | ZAB234        | 145000 |
|          19 | Joseph      | Hernandez | CDE567        | 150000 |
|          20 | Abigail     | Smith     | FGH890        | 155000 |
|         101 | NULL        | NULL      | NULL          | 100001 |
+-------------+-------------+-----------+---------------+--------+
20 rows in set (0.00 sec)

mysql>
```

```
mysql> SELECT * FROM employees;
+-------------+--------------+-----------+---------------+--------+
| employee_id | first_name   | last_name | device_serial | salary |
+-------------+--------------+-----------+---------------+--------+
|           1 | John         | Smith     | ABC123        |  60000 |
|           2 | Jane         | Doe       | DEF456        |  65000 |
|           3 | Bob          | Johnson   | GHI789        |  70000 |
|           4 | Sally        | Fields    | JKL012        |  75000 |
|           5 | Michael      | Smith     | MNO345        |  80000 |
|           6 | Emily        | Jones     | PQR678        |  85000 |
|           7 | David        | Williams  | STU901        |  90000 |
|           8 | Sarah        | Johnson   | VWX234        |  95000 |
|           9 | James        | Brown     | YZA567        | 100000 |
|          10 | Emma         | Miller    | BCD890        | 105000 |
|          11 | William      | Davis     | EFG123        | 110000 |
|          12 | Olivia       | Garcia    | HIJ456        | 115000 |
|          13 | Christopher  | Rodriguez | KLM789        | 120000 |
|          14 | Isabella     | Wilson    | NOP012        | 125000 |
|          15 | Matthew      | Martinez  | QRS345        | 130000 |
|          16 | Sophia       | Anderson  | TUV678        | 135000 |
|          17 | Daniel       | Smith     | WXY901        | 140000 |
|          18 | Mia          | Thomas    | ZAB234        | 145000 |
|          19 | Joseph       | Hernandez | CDE567        | 150000 |
|          20 | Abigail      | Smith     | FGH890        | 155000 |
+-------------+--------------+-----------+---------------+--------+
20 rows in set (0.00 sec)
```

```
mysql> SELECT first_name FROM employees WHERE salary>145000;
+-------------+
| first_name |
+-------------+
| Joseph      |
| Abigail     |
+-------------+
2 rows in set (0.00 sec)

mysql>
```

```
mysql> EXPLAIN SELECT first_name FROM employees WHERE salary>145000;
+----+-------------+-----------+------------+------+---------------+------+---------+------+------+----------+-------------+
| id | select_type | table     | partitions | type | possible_keys | key  | key_len | ref  | rows | filtered | Extra       |
+----+-------------+-----------+------------+------+---------------+------+---------+------+------+----------+-------------+
|  1 | SIMPLE      | employees | NULL       | ALL  | NULL          | NULL | NULL    | NULL |   20 |    33.33 | Using where |
+----+-------------+-----------+------------+------+---------------+------+---------+------+------+----------+-------------+
1 row in set, 1 warning (0.04 sec)

mysql>
```

```
mysql> CREATE index index1 ON employees(salary);
Query OK, 0 rows affected (0.12 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> EXPLAIN SELECT first_name FROM employees WHERE salary>145000;
+----+-------------+-----------+------------+-------+---------------+--------+---------+------+------+----------+-----------------------+
| id | select_type | table     | partitions | type  | possible_keys | key    | key_len | ref  | rows | filtered | Extra                 |
+----+-------------+-----------+------------+-------+---------------+--------+---------+------+------+----------+-----------------------+
|  1 | SIMPLE      | employees | NULL       | range | index1        | index1 | 5       | NULL |    2 |   100.00 | Using index condition |
+----+-------------+-----------+------------+-------+---------------+--------+---------+------+------+----------+-----------------------+
1 row in set, 1 warning (0.00 sec)

mysql> DROP index index1 ON employees;
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT first_name FROM employees WHERE employee_id=10;
+-------------+
| first_name |
+-------------+
| Emma        |
+-------------+
1 row in set (0.00 sec)

mysql>
```

```
mysql> EXPLAIN SELECT first_name FROM employees WHERE employee_id=10;
+----+-------------+-----------+------------+-------+---------------+---------+---------+-------+------+----------+-------+
| id | select_type | table     | partitions | type  | possible_keys | key     | key_len | ref   | rows | filtered | Extra |
+----+-------------+-----------+------------+-------+---------------+---------+---------+-------+------+----------+-------+
|  1 | SIMPLE      | employees | NULL       | const | PRIMARY       | PRIMARY | 4       | const |    1 |   100.00 | NULL  |
+----+-------------+-----------+------------+-------+---------------+---------+---------+-------+------+----------+-------+
1 row in set, 1 warning (0.00 sec)

mysql> CREATE index index1 ON employees(employee_id);
Query OK, 0 rows affected (0.10 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> EXPLAIN SELECT first_name FROM employees WHERE employee_id=10;
+----+-------------+-----------+------------+-------+---------------+---------+---------+-------+------+----------+-------+
| id | select_type | table     | partitions | type  | possible_keys | key     | key_len | ref   | rows | filtered | Extra |
+----+-------------+-----------+------------+-------+---------------+---------+---------+-------+------+----------+-------+
|  1 | SIMPLE      | employees | NULL       | const | PRIMARY,index1 | PRIMARY | 4       | const |    1 |   100.00 | NULL  |
+----+-------------+-----------+------------+-------+---------------+---------+---------+-------+------+----------+-------+
1 row in set, 1 warning (0.00 sec)
```

```
mysql> SHOW index FROM employees WHERE cardinality=20;
+-----------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+-------------
-+---------+------------+
| Table     | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment
 | Visible | Expression |
+-----------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+-------------
-+---------+------------+
| employees |          0 | PRIMARY  |            1 | employee_id | A         |          20 |     NULL |   NULL |      | BTREE      |         |
 | YES     | NULL       |
+-----------+------------+----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+-------------
-+---------+------------+
1 row in set (0.00 sec)

mysql>
```